INVESTOR & STRATEGY BRIEF

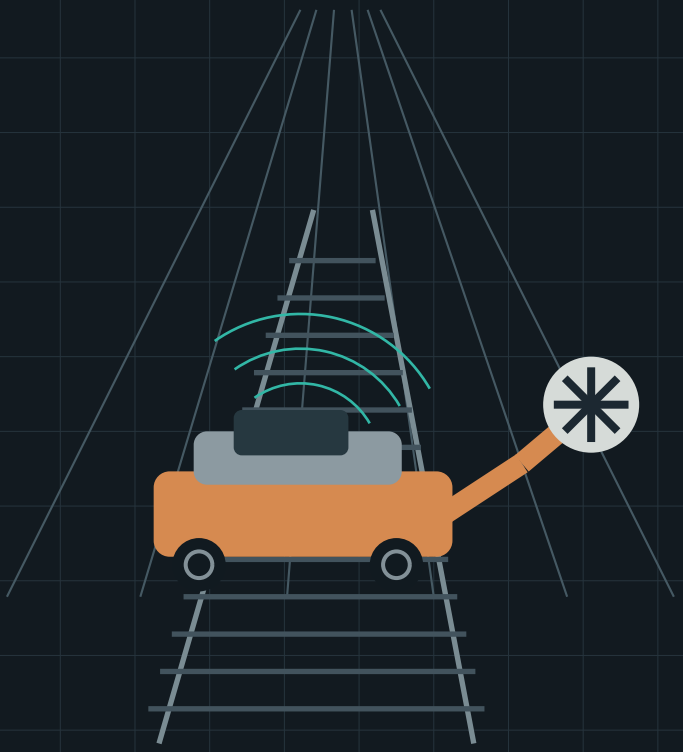
BRITAIN'S COALFIELDS

THE AUTOMATION FRONTIER

A strategic view of UK coal resources, candidate mining regions and the strongest places to trial remote and semi-autonomous systems.

CORE THESIS

The best near-term investment may be the technology layer around the mine - not a new coal reserve. Start where an operating site, a defined customer and a lawful route already exist.



OCTOBER 2026

Speculative commercial concept. Resource geography is not a reserve statement; sites are ranked for diligence, not permission to mine.

INVESTMENT THESIS

Coal remains in the ground. The market has moved.

The UK still has extensive coal-bearing strata and a vast archive of historic workings. That does not translate automatically into an economic reserve: recoverability depends on seam thickness, depth, faulting, gas, water, access, land rights, restoration liabilities, customer specification and planning consent.

28,985 t

UK licensed mine output declared in Q4 2025

MRA quarterly returns

8 sites

Reported sites with output or manpower in Q4 2025

Not eight large operating mines

120,000+

Coal abandonment plans held by MRA

Historic data advantage

WHY THIS IS HARD

UK coal demand has collapsed. DUKES reports overall demand down 96% from 2000 to 2024, and coal-fired electricity generation has ended. New extraction faces a restrictive planning environment, uncertain customer demand and lengthy, capital-intensive permitting.

WHERE VALUE MAY SIT

A credible start-up should sell more than tonnes: safer remote operation, equipment uptime, lower exposure hours, better seam and mine-water intelligence, selective recovery at already permitted sites, and adjacent mine remediation or heat/methane services.

STRATEGIC POSITION

Build the mining-control and data platform first. Use a licensed, operating mine as the proving ground; treat ownership of a new coal asset as a separate, high-risk decision that only follows verified resource, offtake, permissions and financing.

IMPORTANT DISTINCTION

BGS resource maps are regional screening tools, not a bankable reserve report. A resource polygon does not tell an investor how many tonnes can be sold profitably. This brochure identifies candidate regions for technical and commercial diligence; it does not certify reserve quantities, mine economics or project permissions.

Evidence base: MRA Q4 2025 production returns; MRA mine-record guidance; DESNZ DUKES 2025 Chapter 2.

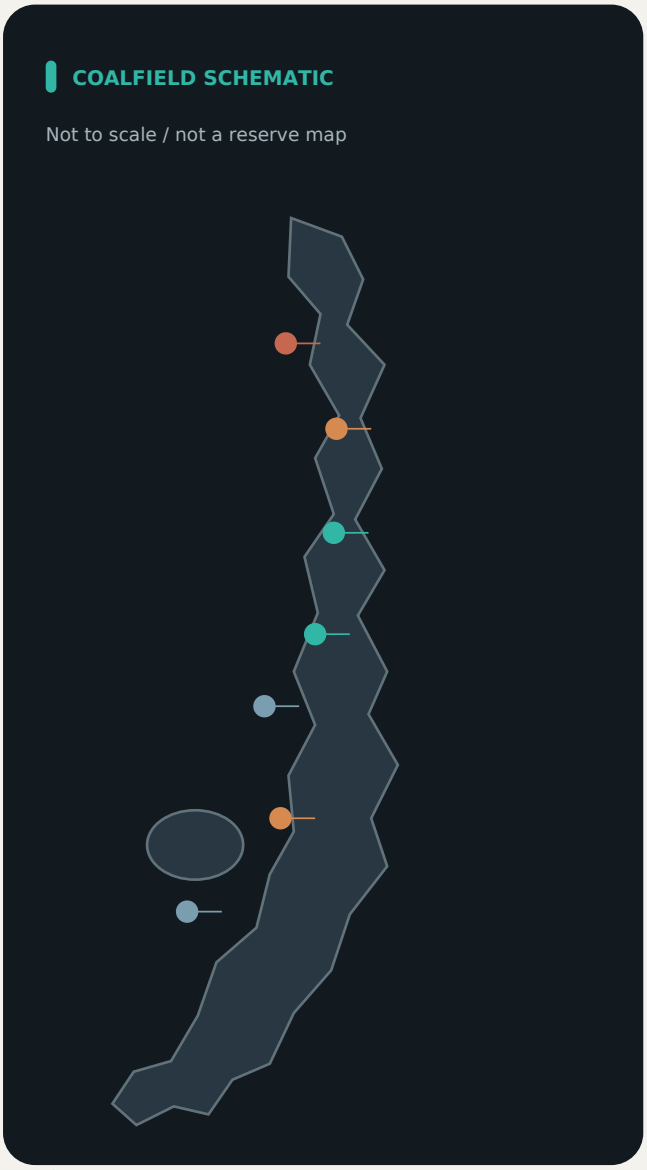
GEOLOGY & RESOURCE

The UK coal map: big basins, uneven economics

Coal-bearing rocks are distributed across Great Britain and parts of Northern Ireland. The major historical basins include South Wales, East Midlands/Yorkshire, Northumberland-Durham, Lancashire, the Scottish Central Belt, West Cumbria, Forest of Dean, Bristol-Somerset and smaller coalfields. Exact rankings change with whether the comparison is geological extent, depth, coal rank or technically recoverable tonnes.

COALFIELD SCHEMATIC

Not to scale / not a reserve map



- Large legacy coalfield
- Existing / specialist operations
- Small-scale / niche
- High policy constraint

SOUTH WALES

High-rank coal and an active licensed operating context. Strongest first candidate for a real underground systems partnership; commercial viability still depends on the seam and buyer.

EAST MIDLANDS / YORKSHIRE

One of the UK's major historic coal provinces. Best for historic data studies, equipment engineering and simulated mine planning; do not infer permission to reopen.

NORTHUMBERLAND-DURHAM

Extensive legacy workings and coal-bearing strata. Screen for subsidence, water, gas and mine-water service opportunities before extraction ideas.

WEST CUMBRIA

A major historic coal region with a contested metallurgical-coal narrative. Treat any new mine as a separate, high-policy-risk development case.

FOREST OF DEAN / AYLE

Small and specialised mining context. Useful for low-throughput sensing and remote operations trials, not a scalable tonnage assumption.

SCOTTISH CENTRAL BELT

Geologically significant legacy coalfield, but Scotland states it does not support coal extraction. Keep work to non-extractive technology, heritage or mine-water services.

RESOURCE BASELINE

BGS maps are regional screening tools (core set dates to 2004), not reserves. Verify with current mine plans, boreholes and independent resource work.

SITE SHORTLIST

Where to test automated mining first

Ranked for a start-up seeking a safe, data-rich proving ground - not ranked by total coal tonnes. Every candidate requires operator agreement, site-specific engineering and regulatory sign-off.

RANK / REGION	PILOT FIT	COMMERCIAL ROLE	MAIN GATE	VIEW
01 SOUTH WALES Aberpergwm operating context	First live underground automation partnership; output also declared in Apr-Jun 2026 returns.	Host access, mine safety case, customer spec, capex/benefit proof.		PRIORITY
02 FOREST OF DEAN Small specialist mines	Micro-pilot for rugged sensors, machine telemetry and operator interface; not a scale mine.	Tiny/variable output, site heterogeneity, logistics.		NARROW
03 AYLE / CUMBRIA Very small licensed operation	Possible targeted tests for sensing, inspection and low-footprint logistics.	Low production scale; geology and access are site-specific.		NARROW
04 EAST MIDLANDS Yorkshire / Notts historic basins	Best regional ecosystem for digital twins, legacy plan analytics, equipment test rigs and engineering talent.	No implied right to reopen; difficult new-extraction policy.		R&D BASE
05 NORTH EAST Deep legacy data footprint	Mine-water, subsidence, methane monitoring and survey technology before any coal proposal.	Old workings, water, faulting and liability.		SERVICES
06 CENTRAL SCOTLAND Historic Central Belt coalfield	Remote inspection / mine heritage / mine-water technology only under current policy stance.	Scottish Government position of no support for coal extraction.		NO NEW MINE

RECOMMENDATION

Secure a memorandum of understanding with an existing licensed mine first. Run a 90-day baseline study and a tightly scoped sensing / remote-operations pilot before building bespoke equipment or buying coal rights.

Candidate sites are indicative. The MRA licence register, mine plans, operating status and land rights must be checked before site selection.

MARKET & UNIT ECONOMICS

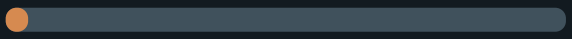
What could actually be profitable?

The strongest commercial thesis is selective, low-volume and customer-led. UK domestic demand is too small and structurally declining to underwrite a speculative high-volume thermal-coal mine. Higher-grade products may command a premium, but only if independently tested and matched to a buyer.

DEMAND RESET

-96%

Overall UK coal demand fell 96% between 2000 and 2024, according to DUKES 2025.



2000 demand index: 100
2024 demand index: 4

REMAINING INDUSTRIAL DEMAND

0.6 Mt

Coal used by UK iron and steel in 2024. Steel represented 27% of UK coal demand that year - a narrower, changing customer base rather than an unlimited market.

FOUR REVENUE LANES TO SCREEN

01 Specialist coal from existing licensed operations

Potentially the nearest route to commodity cash flow. Requires a verified seam, quality certificate, buyer, transport route and low-cost production. No price or margin assumed.

02 Residual recovery at already disturbed sites

May improve yield where historic methods left recoverable material. Screen geotechnical risk, rehabilitation obligation, site consent and true incremental cost first.

03 Automation-as-a-service

Remote operations, machine health, gas/ventilation telemetry, collision awareness and logistics optimisation sold as equipment, software and support. Can monetise without owning coal.

04 Mine legacy services

Mine-plan digitisation, subsidence monitoring, methane capture/management, mine-water monitoring and heat opportunity screening. Often better aligned with the policy exception and legacy assets.

Commercial rule: do not publish “UK reserves” as a single bankable number. Before an investment decision, commission an independent seam-level resource model and a full life-of-mine cost curve with closure security.

PRODUCT STRATEGY

The automation stack: start with the data

Build a retrofit-first system that can run beside incumbent equipment. The first sellable product should improve visibility and reduce exposure time; autonomous cutting should be a later phase, only after a host mine validates the safety case.

05 — OPERATIONS CONTROL

Remote supervisory control, permissions, emergency stop and human-in-the-loop operating rules

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04 — DIGITAL TWIN

Mine geometry, seam model, equipment state, ventilation constraints and production plan

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03 — SENSOR FUSION

Machine telemetry, gas, ventilation, position, vibration, temperature and communications health

↓

02 — MACHINE INTERFACE

Retrofit interfaces to conveyors, cutters, haulage, power systems and maintenance records

↓

01 — DATA FOUNDATION

Georeferenced mine plans, surveys, boreholes, shift logs, incidents and ground-truth labels

PILOT KPIS - PRE-AGREE BASELINE AND ACCEPTANCE

EXPOSURE	UPTIME	SAFETY	OUTPUT
Operator hours in hazardous zone	Unplanned downtime per shift	Near-miss / unsafe-state detection	Tonnes per machine-hour, where relevant

Design principle: monitor first, then advisory control and bounded remote operation - never claim full autonomy before the mine operator signs off.

PERMISSIONS & RISK

A lawful route matters more than a rich seam

Policy is the binding commercial constraint. In England, the National Planning Policy Framework published on 29 September 2026 says development plans should not identify new sites or extensions for coal extraction, and proposals for surface or underground coal working should be refused except for specified purposes such as public safety, ground stabilisation, incidental extraction or methane capture. Scotland has stated a position of no support for coal extraction.

PLANNING / LICENCE

Separate the tech pilot from any new extraction proposal. Verify current licensing policy, mineral rights, planning consent and environmental permits before spend.

High

RESOURCE / GEOLOGY

Commission independent seam mapping, core analysis, gas/water assessment, fault mapping and a mine-plan reconciliation. Regional maps are only screening tools.

High

OFFTAKE / PRICE

Get written product specs and conditional offtake interest before assuming coal revenue. Model import parity, haulage, washing, royalties and restoration.

High

SAFETY / LIABILITY

Pilot under the mine operator's safety management system; quantify ground control, methane, water, ventilation, communications and emergency shut-down risks.

High

CAPITAL / CLOSURE

Use milestones, capped pilot exposure, insurance, closure/rehabilitation provision and appropriate financial security. Avoid greenfield fixed costs until evidence improves.

Medium-high

INVESTMENT GATES

0 Desk screen

1 Site + data access

2 Safety / baseline

3 Pilot evidence

4 Commercial scale

No gate advances on a geological map alone. Each requires named owner, written evidence, and a board-approved stop/go decision.

NEXT STEPS

The 12-month plan and sourcebook

A low-regret launch plan validates the customer and the machine interface before it validates a new coal asset.

FIRST YEAR ROADMAP

MONTH 0-2

Choose host + scope

Pick one incumbent operator; secure data access; agree what success looks like.

MONTH 2-4

Mine-data baseline

Digitise available plans; hazard / network survey; reliability and exposure baseline.

MONTH 4-7

Bench / surface trial

Test sensors, rugged interfaces, communications resilience and fail-safe behaviour.

MONTH 7-10

Underground bounded pilot

Run advisory monitoring or remote inspection with operator approval and supervised testing.

MONTH 10-12

Prove unit economics

Compare against baseline; publish verified KPI delta; decide to expand, redesign or stop.

BOARD DECISION FRAME

Proceed only if:

- An operating mine agrees to host a bounded pilot.
- Data access and confidentiality terms are signed.
- A safety case and rollback plan are accepted by the operator.
- An independent engineer validates the performance baseline.
- A paying customer or credible buyer is identified.
- Pilot capex is within a pre-agreed loss limit.
- Coal asset acquisition is optional, not a precondition.

THE THESIS

Use UK coalfields as a first proving ground, then sell the platform into wider underground-mining and mine-remediation markets.

SOURCEBOOK / CHECKED 13 OCTOBER 2026

1. BGS - Coal resources for new technologies (regional screening maps; original maps published 2004).
2. MRA - Coal mining production and manpower returns, Oct-Dec 2025 (official declared output).
3. MRA - Coal mining production and manpower returns, Apr-Jun 2026 (latest quarterly site returns available at publication).
4. MRA - Access coal mining information and data (mine plans, licences and mapping).
5. MRA - Guidance notes for surface coal mining licences.
6. MHCLG - NPPF Chapter 11: coal decision policies, updated 29 Sep 2026.
7. Scottish Government - Coal extraction policy / notification direction, June 2025.
8. DESNZ - DUKES 2025 Chapter 2; coal demand trend and 2024 iron/steel consumption.
9. BGS / National Data Library - UK Coal Resource for New Exploitation Technologies dataset summary.

Disclaimer: this is a speculative strategy brochure, not investment advice, a reserve statement, a feasibility study or legal advice. Output figures are reported declarations; site availability and legal position must be refreshed at the point of diligence.